Lower body specialization — Session B (Wednesday)

Quad-supported emphasis · hypermobility-safe · knee rehab phase · ~52 min · 25 working sets

Muscle group

Quads · Hamstrings · Glutes · Calves

30–60 min before: collagen peptides (10–15g) + vitamin C
Maximizes collagen synthesis in tendons loaded during session

B — Superset 1: Quad + Knee flexion (~9 min · 3 rounds · 90s rest after each pair)

1 **Smith machine box squat (bench at 90° knee)** — 3 × 8–10
Quads (dominant) · glutes (synergist) · primary quad compound — Smith · 3s ecc
Bilateral · controlled depth to bench · 3s lowering · soft lockout

↕ then immediately

2 **Prone hamstring curl** — 3 × 10–12
Hamstrings — knee flexion — Band / light DB
Strict antagonist to box squat · light load · no knee snap

C — Superset 2: Quad + Adductor iso (~8 min · 3 rounds · 90s rest after each pair)

3 **Goblet box squat (DB at chest)** — 3 × 10–12
Quads (dominant) · 2nd quad compound — DB
Bilateral · controlled depth · soft lockout at top

↕ then immediately

4 **Copenhagen plank holds (bench-supported)** — 2 × 15s/side
(top leg on bench · bottom leg suspended · 20–30% effort) — Bodyweight
Adductors · isometric · cross-region pairing — zero overlap with squat

D — Superset 3: Glute iso + Calf (~7 min · 3 rounds · 90s rest after each pair)

5 **Banded glute bridge isometric hold** — 3 × 20s
Glutes · hamstrings (iso stabilizer) — Band
Drive through heels · stop short of full hip extension · squeeze and hold

↕ then immediately

6 **Standing calf raises (Smith · optional deficit)** — 3 × 12–15
(3s eccentric · 1s pause at bottom stretch · soft knee) — Smith
Gastrocnemius (primary) · soleus synergist
Optional deficit: plate under balls of feet · heel drop ~3–5cm · 70–80% range

E — Finisher: Abductor + Tibialis (~4 min · straight sets · 45s rest)

7 **Side-lying banded hip abduction** — 3 × 12/side
Glute medius · hip abductors — Band
Knee stabilizer maintenance

8 **Tibialis raises (wall-supported)** — 4 × 15
Tibialis anterior · ankle dorsiflexion strength — Bodyweight / band
Lean back against wall · lift toes against bodyweight or band · landing/badminton prep

Within 60 min after: protein 30–40g + vitamin C
Supports muscle + tendon remodeling · aim 1.6–2g protein/kg/day total

Every rep, every set — hypermobility + knee rehab rules

- Never lock out knees — soft bend at end range
- No single-leg eccentric loading through knee flexion-extension cycles (rehab phase)
- Skip session if joints feel unusually loose that day — reschedule, don't push
- Stop set early if any joint feels unstable — prioritize joint feel over rep count
- No pain rule — stop immediately if any knee pain or swelling
- Lengthened-position work targets ~70–80% of available range — never end range
- Hip abductor/adductor machine: deferred until knee stabilizes

Session timeline

Superset B (9 min) → Superset C (8 min) → Superset D (7 min) → Finisher E (4 min)
Total working session: ~52 min · 25 working sets · then proceed to post-workout mobility protocol

Working set count — Session B

Quads: 6 sets (Smith box 3 + goblet box 3)
Hamstrings: 3 sets (prone curl)
Glutes: 3 sets (banded bridge iso)
Calves: 3 sets (standing)
Hip abductors: 3 sets (side-lying)
Adductors: 2 sets (Copenhagen iso)
Tibialis ant: 4 sets

Alternative exercises — if Smith machine occupied

# Primary exercise	Alternative 1	Alternative 2
1 Smith machine box squat	Barbell box squat (free bar on back)	Heel-elevated goblet squat
6 Standing calf raises (Smith)	DB standing calf raise (hold rack)	Single-leg DB calf raise (hold rack)

v3 — warmup removed (redundant with mobility doc) · alts audit (row 1 reorder · row 6 BB calf raise → single-leg DB)